Dr. S. Balaji, Ph.D. | Mathematics

SASTRA Deemed University – Thanjavur, Tamil Nadu 613401 – India
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balajimath.github.io
ORCiD Scopus Google Scholar University Profile

PROFESSIONAL SUMMARY

Distinguished mathematician with **25 years of progressive academic experience** and proven track record in research, teaching, and academic leadership. Working to leverage expertise in **algorithmic graph theory, combinatorial optimization, numerical methods, and wavelet approximation** while continuing to mentor doctoral students and contribute to institutional excellence.

Research Excellence

Publications	SCOPUS	h-index/i10-index	Citations	IF/SNIP
39 Papers	39	17/23	965	30.66/20.04

Authorship	Corresponding	First	Other	Sole
Percentage	85%	10%	5%	10 Papers

Specializations: • Algorithmic Graph Theory • Combinatorial Optimization • Numerical Methods • Wavelet Approximation

Key Publications

2026: Devi, V.,**Balaji, S.** A Hybrid Jumping Particle Swarm Optimization with Local Search for Airline Crew Pairing: Theoretical and Empirical Insights," *National Academy of Science Letters*, (Article in Press), https://doi.org/10.1007/s40009-026-02103-z.. **IF: 1.4**

2025: Maheswari, G.,**Balaji, S.** "A hybrid jumping swarm optimization based local search approach for minimum weighted dominating set problem," *Evolving Systems*, 16:1-15. **IF: 2.7**

2024: **Balaji, S.,** Hariharan, G., "An efficient wavelet-based approximation method for solving nonlinear fractional-time long wave equations: An operational matrix approach," *Mathematical Methods in the Applied Sciences*, 47:1015-1033. **IF: 2.9**

2021: **Balaji, S.**, Vikram, S.T., Kanagasabapathy, G., "Jumping particle swarm optimisation method for solving minimum weight vertex cover problem," *International Journal of Bio-Inspired Computation*, 18(3):143-152. **IF: 3.5**

2016: **Balaji, S.**, "A new Bernoulli wavelet operational matrix of derivative method for the solution of nonlinear singular Lane-Emden type equations arising in astrophysics," *Journal of Computational and Nonlinear Dynamics (ASME)*, 11(051013). **IF: 1.8**

2014: **Balaji, S.**, "A new approach for solving Duffing equations involving both integral and non-integral forcing terms, " *Ain shams engineering journal*, 5(3), 985-990. **IF: 5.9**

Education

SASTRA Deemed University — **Thanjavur**
Ph.D. in Mathematics — *2005–2012*
Thesis: "Effective Heuristics on Graph Optimization Problems"
Selected in the pool of best theses

Madurai Kamaraj University — **Madurai**
M.Phil. Mathematics, First Class — *2003–2004*

The MDT Hindu College — **Tirunelveli**
M.Sc. Mathematics, Distinction — *1999–2001*
College Topper Award

Sri Paramakalyani College — **Alwarkuruchi**
B.Sc. Mathematics, First Class — *1996–1999*

Academic Experience

SASTRA Deemed University — **Thanjavur**
Lecturer - Professor — *Sept 2004–Present*
21 Years of Continuous Service

- Teaching UG (B.Tech/B.Sc) and PG (M.Tech/M.Sc) level mathematics courses
- **Doctoral Supervision:** 2 Ph.D. awarded, 2 currently supervising
- Research focus: Graph optimization, combinatorial algorithms, numerical methods
- Active contributor to NAAC & NBA accreditation processes
- Board member of syllabus revision committee
- Designed AI based syllabus for various programmes
- Course coordinator for online degree programs (BCA, M.Com)

Sri Paramakalyani College — **Alwarkuruchi**
Lecturer — *June 2003–Aug 2004*

- Managed B.Sc courses for regular and evening batches
- Provided additional academic support for struggling students

The MDT Hindu College — **Tirunelveli**
Lecturer — *Aug 2001–Apr 2003*

- Taught B.Sc and M.Sc level mathematics courses

Doctoral Supervision & Mentoring

Ph.D. Graduated: Two scholars graduated

- **Dr. S.T. Vikram** (2024) – "Strong Chromatic index of graphs derived from Path, Star, and Cycle graphs"
- *Current Position:* Assistant Professor (SG), Rajalakshmi College of Engineering, Chennai.
- **Dr. N. Revathi** (2025) – "A Computational Study On Jumping Particle Swarm-Based Heuristics For Selected Graph Optimizations Problems"
- *Current Position:* Assistant Professor, The Marudhu Pandiyar College, Thanjavur.

Ph.D. In Progress: Two in Progress and one going to submit thesis

- **G. Maheswari** – Algorithmic Graph Theory
- **V. Devi** – Algorithmic Graph Theory

M.Sc. Projects: Supervised **20 M.Sc. Data Science projects** (2020-2025) and **25 Integrated M.Sc. Mathematics projects** (2015-2025) in areas including:

- AI-driven applications (Drug-Target Interactions, Traffic Accident Prevention)
- Machine Learning (Spam Detection, Fake News Detection, Alzheimer's Prediction, Weather Forecast)
- Agricultural Data Science (Agroclimatic Clustering for Crop Yield)
- Graph Optimization, Combinatorial Optimization, Numerical Methods, Wavelet Approximations

Research Funding

Principal Investigator: **Project:** "Vertex Cover-based Link Monitoring and Backbone Formation in Wireless Ad hoc Networks"
Funding Agency: SASTRA-TRR Research Fund
Duration: Feb 2024 – Feb 2025
Amount: ₹ 1,59,000
Status: Completed successfully with three SCIE Journal publications

Awards & Recognition

2023: **Best Paper Award** – IEEE 9th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore

2018: **Certificate of Reviewing** – Journal of Computational and Applied Mathematics, Elsevier

2012: **Cash Prize Award** – ₹ 1,00,000 for timely Ph.D. completion, SASTRA University

2001: **Merit Scholarship** – College topper with distinction in M.Sc., MDT Hindu College

Academic Leadership & Service

Editorial & Review Service

- Peer Reviewer for Elsevier, Springer, Taylor & Francis, Hindawi
- Ph.D. External Examiner – Bharathiyar University, Coimbatore

Conference Organization

2020: **Webinar Coordinator** – "Mathematical Modelling and Its Applications" (9-lecture series with IIT, IISc, NIT, IMSc speakers)

2016: **Session Chair** – 19th Ramanujan Symposium, RIASM, University of Madras

Institutional Service

- NAAC & NBA accreditation coordinator
- Curriculum development for UG/PG programs
- SWAYAM-NPTEL video content creation
- Student admissions and counseling

Teaching Excellence

Academic Excellence

Student Feedback: Consistently maintaining feedback score $\geq$ **4.65 to 5** / 5

Student Engagements: Mentoring, Bridge Courses, & Coaching Classes

Course Development

Online: **MATOL120** – Inferential Discrete Mathematics (BCA program)

MCOMOL104 – Advanced Business Statistics (M.Com program)

Book Contributions

2023: Chapter: **"Vector Differentiation and Integration"** in *Mathematics for Engineers*, Pearson India

2023: Tamil Translation: **"Thisaiyan Nunganitham"**, Pearson India

2014: Chapter: **"Purpose of Evaluation"** in *SASTRA B.Ed Course Materials*, Pearson India

Invited Lectures & Presentations

Jan 2026: **One-Day Symposium** – "AI as Teaching Assistant - Practical Experiments in AI-Enhanced Pedagogy" SASTRA Deemed University (Second Teaching &LC Symposium)

July 2025: **One-Day Workshop** – "AI in Mathematics – Ideas to Experiment in 2025-26" SASTRA Deemed University (Synergistic Learning of Mathematics for Engineering Programme Students)

Dec 2024: **National Mathematics Day** – Bon Secours College for Women, Thanjavur
Theme: "Mathematics: The Bridge to Innovation and Progress"

Jan 2024: **5-Day Workshop** – "Transforming Science through Data-Driven Models" SASTRA Deemed University (Scientific Machine Learning in Ocean Engineering)

July 2024: **IEEE ICEEICT 2024** – "High-Performance Digital Filters in Electronic Stethoscope" Third International Conference, Trichy

Feb 2019: **National Conference** – "Algorithmic Graph Theory" Marudupandiyar College, Thanjavur

Mar 2016: **Ramanujan Institute (RIASM)** – "Bernoulli Wavelet operational matrix of derivatives" University of Madras, Chennai

Professional Development

2025: **AI in Industrial & Management Engineering** – NPTEL/IIT Kanpur | **Grade: Elite + Silver**

2025: **NPTEL-IIT Kanpur FDP** – AI in Industrial & Management Engineering

2024: **ChatGPT & AI Hacks with MS Office** – Microsoft Certified Training

2020: **LaTeX for Scientists and Engineers** – IIT Bombay | **Grade: A+**

2021: **AICTE-ATAL FDP** – Financial Mathematics & Machine Learning

2021: **AICTE-ATAL FDP** – Big Data and Analytics

Technical Skills

Programming: LaTeX, MATLAB, C, C++

Research Tools: Mathematical modeling, Algorithmic design, Data analysis

Emerging Tech: AI/ML applications, Data Science integration

VISION

I aim to: **(1)** Expand research impact through collaborative networks and external funding, **(2)** Increase doctoral supervision capacity and research output, **(3)** Integrate AI and data science with traditional mathematics curricula, **(4)** Bridge academic research with industry applications, and **(5)** Develop international research partnerships in algorithmic mathematics.

LIST OF PUBLICATIONS

Dr. **S. Balaji**

2026

- V. Devi, **S. Balaji**
  A Hybrid Jumping Particle Swarm Optimization with Local Search for Airline Crew Pairing: Theoretical and Empirical Insights.
  National Academy of Science Letters, (Article in Press), https://doi.org/10.1007/s40009-026-02103-z.
- G. Maheswari, **S. Balaji**
  A Jumping swarm optimization-based hybrid approach for solving the minimum dominating set problem.
  2025 IEEE 2nd International Conference for Women in Computing (InCoWoCo), India, pp. 1-10, doi: 10.1109/InCoWoCo68239.2025.11407240.

2025

- G. Maheswari, **S. Balaji**
  A hybrid jumping swarm optimization based local search approach for minimum weighted dominating set problem.
  Evolving Systems, Vol. 16, pp. 1–15.
- G. Maheswari, **S. Balaji**
  A Swarm Intelligence Technique for Link Monitoring Problems in Wireless Ad Hoc Networks.
  National Academy Science Letters, Vol. 48, pp. 613–618.
- G. Maheswari, **S. Balaji**, N. Revathi, R. Nithyakala
  Jumping particle swarm-based local search approach for the maximum weighted independent set problem.
  Discrete Mathematics, Algorithms and Applications, 2550063.

2024

- **S. Balaji**, M. Subbarao, S. Kiruthik, V. J. M. Jerard, P. Ganesan, V. Elamaran
  High-Performance Digital Filters in an Electronic Stethoscope: An FPGA Based Approach.
  2024 Third International Conference on Electrical, Electronics, Information and Communication Technologies (ICEEICT), IEEE, pp. 1–7.
- **S. Balaji**, G. Hariharan
  An efficient wavelet-based approximation method for solving nonlinear fractional-time long wave equations: An operational matrix approach.
  Mathematical Methods in the Applied Sciences, Vol. 47, pp. 1015–1033.
- V. S. Thiru, **S. Balaji**
  Strong chromatic indices of certain binary operations on graphs.
  Discrete Mathematics, Algorithms and Applications, Vol. 16(06), 2350073.

2023

- S. T. Vikram, **S. Balaji**
  An Analysis of the Factors Influencing the Strong Chromatic Index of Graphs Derived by Inflating a Few Common Classes of Graphs.
  Symmetry, Vol. 15, 1301.
- N. Revathi, R. Nithyakala, **S. Balaji**
  A Bio-Inspired Technique for the Maximum Weighted Clique Problem.
  National Academy of Science Letters, Vol. 46(3), pp. 223–227.
- S. Fairooz, **S. Balaji**, R. Ramya, P. Thanapal, V. Elamaran
  A Case Study using Simple Moving Average Filters to accomplish ECG denoising on an FPGA.
  9th International Conference on Advanced Computing and Communication Systems (ICACCS 2023), IEEE, pp. 244–248.
- T. R. Reddy, **S. Balaji**, R. Ramya, P. Thanapal, V. Elamaran
  Analyzing Data Compression Techniques for Biomedical Signals and Images Using Down-sampling and Upsampling.
  9th International Conference on Advanced Computing and Communication Systems (ICACCS 2023), IEEE, pp. 71–76. (Best Paper Award)

2022

- **S. Balaji**, S. T. Vikram
  Strong chromatic index of generalized Jahangir graphs and generalized Helm graphs.
  Discrete Mathematics, Algorithms and Applications, Vol. 14, 2250045.

2021

- **S. Balaji**, S. T. Vikram
  Strong Chromatic index of generalized polygon snake graphs.
  National Academy of Science Letters, Vol. 44(2), pp. 151–154.
- **S. Balaji**, S. T. Vikram, G. Kanagasabapathy
  Jumping particle swarm optimisation method for solving minimum weight vertex cover problem.
  International Journal of Bio-Inspired Computation, Vol. 18(3), pp. 143–152.

2020

- **S. Balaji**, K. Naganthran, A. Zeeshan, Md. Basir, N. Shehzad, R. Nazar, R. Choudary
  Concentration Flux Dependent on Radiative MHD Casson Flow with Arrhenius Activation Energy: Homotopy Analysis Method (HAM) with an Evolutionary Algorithm.
  International Journal of Heat and Technology, Vol. 38(4), pp. 785–793.
- **S. Balaji**, N. Rajagopal, R. Seethalakshmi, V. S. Balaji
  A new numerical method for fractional order Volterra integro-differential equations.
  Ain Shams Engineering Journal, Vol. 11, pp. 171–177.

2019

- **S. Balaji**, G. Hariharan
  An efficient operational matrix method for the numerical solutions of the fractional Bagley–Torvik equation using wavelets.
  Journal of Mathematical Chemistry, Vol. 57, pp. 1885–1901.

2018

- **S. Balaji**, V. Venkataraman, D. R. V. S. R. K. Sastry
  Thermal and solutal dispersion effects on heat and mass transfer in a viscous MHD flow over a vertical plate.
  International Journal of Pure and Applied Mathematics, Special Issue.
- **S. Balaji**, G. Hariharan
  A novel wavelet approximation method for the solution of nonlinear differential equations with variable coefficients arising in astrophysics.
  Astrophysics and Space Science, Vol. 363(16).

2017

- **S. Balaji**, V. Venkataraman, D. R. V. S. R. K. Sastry
  Hydrodynamic marangoni convection in a nanofluid flow with soret and dufour effects.
  International Journal of Civil Engineering and Technology, Vol. 8, pp. 164–171.
- **S. Balaji**, V. Venkataraman, D. R. V. S. R. K. Sastry
  Solution of system of nonlinear equations using integrated RADM and ADM.
  International Journal of Pure and Applied Mathematics, Vol. 117, pp. 367–373.

2016

- N. Revathi, **S. Balaji**
  A new approach for solving set covering problem using Jumping particle swarm optimization method.
  Natural Computing (Springer), Vol. 15, pp. 503–517.
- **S. Balaji**
  A new Bernoulli wavelet operational matrix of derivative method for the solution of nonlinear singular Lane–Emden type equations arising in astrophysics.
  Journal of Computational and Nonlinear Dynamics (ASME), Vol. 11(051013).
- **S. Balaji**, V. Venkataraman, D. R. V. S. R. K. Sastry
  Effect of radiation on mhd mixed convection flow past a semi infinite vertical plate.
  ARPN Journal of Engineering and Applied Sciences, Vol. 11, pp. 14033–14038.

2015

- **S. Balaji**
  Legendre wavelet method for solution of nonlinear Riccati differential equation.
  Journal of Egyptian Mathematical Society, Vol. 23, pp. 263–270.

2014

- **S. Balaji**
  A new approach for the solution of duffing equation involving both integral and non-integral forcing terms.
  Ain Shams Engineering Journal, Vol. 5, pp. 985–990.
- **S. Balaji**
  Solution of nonlinear Riccati differential equation using Chebyshev wavelets.
  WSEAS Transaction on Mathematics, Vol. 13, pp. 441–451.
- **S. Balaji**
  Legendre wavelet method for solution of nonlinear Riccati differential equation of fractional order.
  International Journal of Mathematics and Mathematical Sciences, Vol. 10, 304745.

2013

- **S. Balaji**, Y. B. Venkatakrishnan, K. Kannan
  Total dominating set based algorithm for minimum connected dominating set in ad hoc wireless networks.
  WSEAS Transaction on Mathematics, Vol. 12, pp. 1164–1172.
- **S. Balaji**
  A local search based greedy heuristic for the minimum vertex cover problem.
  Far East Journal of Mathematical Sciences, Vol. 79, pp. 257–266.
- **S. Balaji**
  A new effective local search heuristic for the maximum clique problem.
  International Journal of Mathematical Sciences, Vol. 7, pp. 32–38.

2012

- **S. Balaji**, N. Revathi
  An efficient approach for the optimization version of the maximum weight clique problem.
  WSEAS Transaction on Mathematics, Vol. 11, pp. 773–783.
- **S. Balaji**, N. Revathi
  An efficient heuristic for the minimum connected dominating set on ad hoc wireless network.
  International Journal of Computer and Communication Engineering, Vol. 6, pp. 278–284.

2010

- **S. Balaji**, V. Swaminathan, K. Kannan
  A Simple Algorithm for Maximum Clique and Matching Protein Structures.
  International Journal of Combinatorial Optimization Problems and Informatics, Vol. 1, pp. 2–11.
- **S. Balaji**, V. Swaminathan, K. Kannan
  Optimization of Unweighted Minimum Vertex Cover.
  Proceedings of World Academy of Science Engineering and Technology, Vol. 67, pp. 508–513.
- **S. Balaji**, V. Swaminathan, K. Kannan
  An Effective Algorithm for minimum weighted vertex cover problem.

International Journal of Computational and Mathematical Sciences, Vol. 4, pp. 34–38.

- **S. Balaji**, V. Swaminathan, K. Kannan
  A Simple Algorithm to Optimize Maximum Independent Set.
  Advanced Modeling and Optimization, Vol. 12, pp. 107–118.

2009

- **S. Balaji**, V. Swaminathan, K. Kannan
  Approximating Maximum Weighted Independent Set Using Vertex Support.
  International Journal of Computational and Mathematical Sciences, Vol. 3, pp. 412–417.

Summary

Total Publications: 37
SCI/SCIE Journals: 18 — SCOPUS/WOS Journals: 37
Webpage Links:
ORCID | Scopus | Google Scholar | University Profile

My Progress in AI Usage

A Continuing Journey of Learning, Creating & Innovating

Apps & Software Tools

Developed **three application/software tools** (mobile/web) using Claude-AI:

- **SASTRA Mentor Connect APP** — an easy-to-use student mentoring platform.
- **Resume Analyzer APP** — compares a student's résumé against a Job Description (JD) and highlights missing keywords/skills.
- **Personal website** — A web-platform displays a whole journey of my academic and research progress.

FDP & Course Certificates

- **Feb 2024:** Completed AI workshop on *ChatGPT & AI Hacks with MS Office*, conducted by Mr. Jatan Shah — Microsoft Certified Trainer, Co-founder & CEO, Skill Nation.
- **Aug–Oct 2025:** Completed **NPTEL-AICTE Faculty Development Programme** for the course *Artificial Intelligence in Industrial and Management Engineering*, conducted by **IIT Kanpur**.
- **Aug–Oct 2025:** Secured **Elite + Silver Grade** in the proctored exam of *Artificial Intelligence in Industrial and Management Engineering*, IIT Kanpur.

Advanced AI Curriculum

- **July 2025:** Framed an AI-enhanced curriculum for the *B.Tech (ECE) V Semester* course — **Probability and Random Processes with AI Integration**.
- **Dec 2025:** Framed an AI-enhanced curriculum for *B.Sc., M.Sc. Mathematics and Data Science* courses under SASTRA University — Zoho initiative of **Building Mathematical Foundations for Modern AI**.

Invited Lectures

- **Jan 2025:** Presented a lecture and hands-on session titled *"Transforming Science through Data-Driven Models"* at the **Five-Day Workshop on Scientific Machine Learning in Ocean Engineering**, SASTRA University.
- **July 2025:** Presented a lecture on *"AI in Mathematics — Ideas to Experiment in 2025–26"* at the **Workshop on Synergistic Learning of Mathematics for Engineering Programme Students**, SASTRA University.

- **Jan 2026:** Presented a lecture titled *"AI as Teaching Assistant — Practical Experiments in AI-Enhanced Pedagogy"* at the **Second TLC Symposium**, SASTRA University.

Research Papers

Three papers with the following titles are **under review** in **SCIE Journals**:

- **An AI-Driven Interactive Interface for Interview Excellence**
- **Rainfall Forecasting Using Machine Learning and Time Series Models**
- **Enhanced Rainfall Prediction using Machine Learning and Actionable Insights**

Academic Activities

- Use of AI tools such as **Google Colab**, **Desmos**, etc., for classroom teaching.
- Use of AI tools such as **Keptune**, **ChatGPT**, **Claude**, etc., for School Result Analyses.

. . . and the journey is *still counting* . . .

(Signature)
Dr. S. Balaji